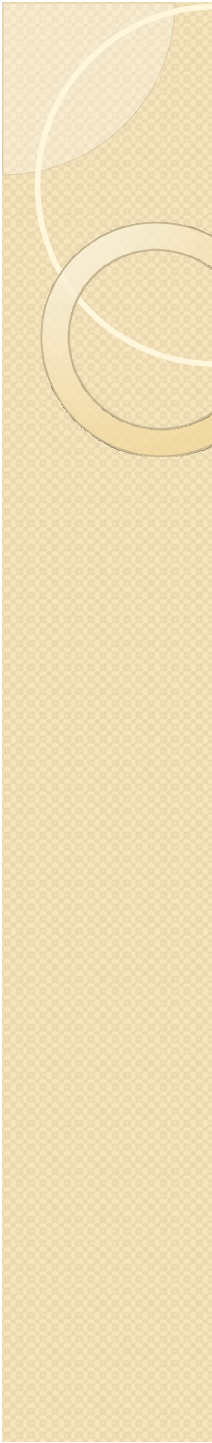


MODULE 3 — Transformers & LLM

Session :8

Why Transformers Changed Everything



SLIDE 1:- Why Transformers Changed Everything

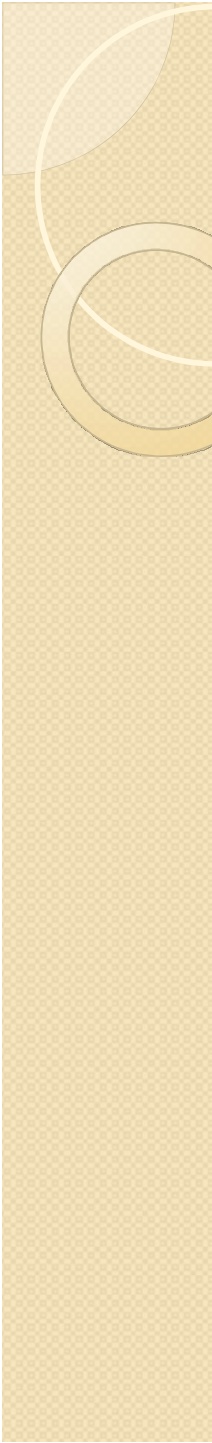
From Sequential Memory to Attention

RNN → Attention → Self-Attention → Transformers → LLMs

Question for Students

How can ChatGPT understand a 10-page document and connect information from different parts?

Today's session answers this question.



SLIDE 2 — The Big Hook

Can You Remember This?

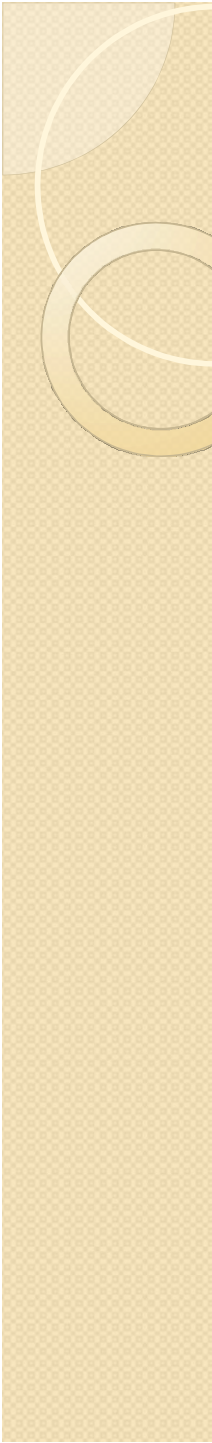
"The student who joined the AI Bootcamp in July and completed Python, Machine Learning, Deep Learning and NLP sessions successfully submitted his final project yesterday."

Q- What did the student submit?

Q- Who submitted the final project?

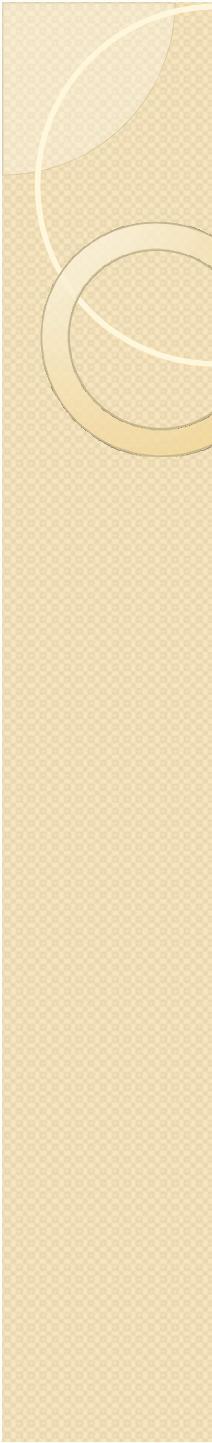
To understand a sentence, we often need to connect information that is **far apart**.

This was difficult for older sequential models.



Students: **Final project**

Students: **The student**



5 students to stand in a line.

Take one word each:

The → student → from → BCA → succeeded

The first student passes information to the second.

Each student can only communicate with the next student.

What was the first word?

They may forget or distort information.

Important Point

This is similar to a **sequential information flow**.

Imagine you could directly look at every word.
That is closer to the main idea behind **Attention**.

SLIDE 3 — Before Transformers

How Did Earlier Language Models Process Text?

One popular approach was:

RNN — Recurrent Neural Network

I → love → learning → Artificial → Intelligence

↓

Previous information flows forward

At every step:

Current Word + Previous Memory

↓

Next Hidden State

Important Idea

RNN processes text one step at a time.

SLIDE 4 — What Is Sequential Processing?

Example

Sentence:

I love learning Artificial Intelligence

An RNN processes:

Step 1 → I

Step 2 → love

Step 3 → learning

Step 4 → Artificial

Step 5 → Intelligence

Question

Can Step 5 happen before Step 4?

✗ No.

RNN must wait.

ACTIVITY 2 — Human RNN

Students sit in a row.

Given sentence:

“Artificial Intelligence will transform the future of software development.”

Rules

Student 1 reads the first word.

Student 2 receives:

Word 2 + Memory from Student 1

Student 3 receives:

Word 3 + Memory from Student 2

Continue.

Ask:

What happens if Student 5 makes a mistake?

Answer:

The incorrect information can affect later students.

Key Point

Sequential processing can create
information bottlenecks.

SLIDE 5 — Problem 1: Long-Term Dependencies

Consider This Sentence

"The AI student who attended the bootcamp for three months and completed many projects ... was successful."

Ask:

Who was successful?

Answer:

The AI student

But "AI student" and "was successful" are far apart.

The model needs to remember important information over a long distance.

This is called a:

Long-Term Dependency

ACTIVITY 3 — Find the Connection

"The girl who studied Python, Java, Machine Learning, Deep Learning, NLP and Transformers for several months became an AI developer."

Things to connect:

girl —————→ AI developer

Q- How many words are between these related concepts?

Explain:

Older sequential models may struggle when important information is far apart.

SLIDE 6 — Problems With RNNs

1. Sequential Processing

Must process:

Word 1 → Word 2 → Word 3 → Word 4

Cannot easily process all words simultaneously.

2. Long-Term Dependency

Information from earlier words may become weaker over long sequences.

3. Slow Training

For a sentence with 100 words:

Step 1



Step 2



Step 3

...



Step 100

The next computation depends on the previous one.

4. Information Bottleneck

Important information has to travel through hidden states.

SLIDE 7 — RNN Visualization

Word 1 → [Memory] → Word 2 → [Memory] → Word 3 →
[Memory] → Word 4

Key Question

What if Word 4 needs information directly from Word 1?

RNN:

Word 1 → Word 2 → Word 3 → Word 4

It travels through everything.

This is inefficient.

SLIDE 8 — The Solution: Attention

What If the Model Could Look Directly at Important Words?

Instead of:

Word 1 $\rightarrow$ Word 2 $\rightarrow$ Word 3 $\rightarrow$ Word 4

Imagine:

Word 4 $\longrightarrow$ Word 1

└── $\rightarrow$ Word 2

└── $\rightarrow$ Word 3

The model decides:

Which information should I focus on?

This is the basic intuition behind:

ATTENTION

SLIDE 9 — Human Attention Example

Read this question:

“Shashank Sir teaches Artificial Intelligence to BCA students.”

Question:

Who teaches AI?

Your brain focuses mainly on:

Shashank sir

It does not give equal importance to every word.

Question:

Who are being taught?

Your focus changes to:

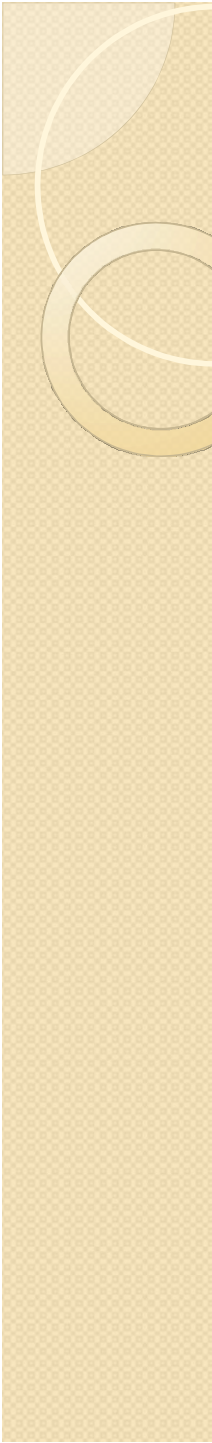
BCA students

Teaching Point

The question/context changes what we pay attention to.

AI Attention works similarly:

Focus more on relevant information.



ACTIVITY 4 — Attention Game

"The AI engineer built a powerful model using a large dataset in the research lab."

Question 1

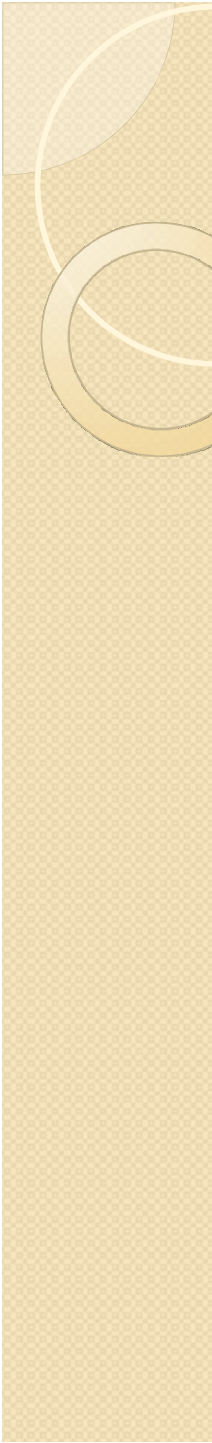
Who built the model?

Question 2

What was used to build the model?

Question 3

Where was it built?



Question 1

Who built the model?

Students highlight:

AI engineer

Question 2

What was used to build the model?

Students highlight:

large dataset

Question 3

Where was it built?

Students highlight:

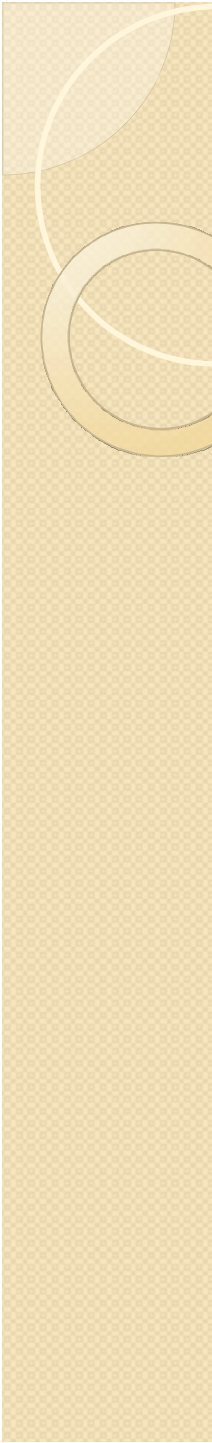
research lab

Important Point

Same sentence.

Different question.

Different attention.



SLIDE 10 — Attention Scores

The model does not simply say:

Important

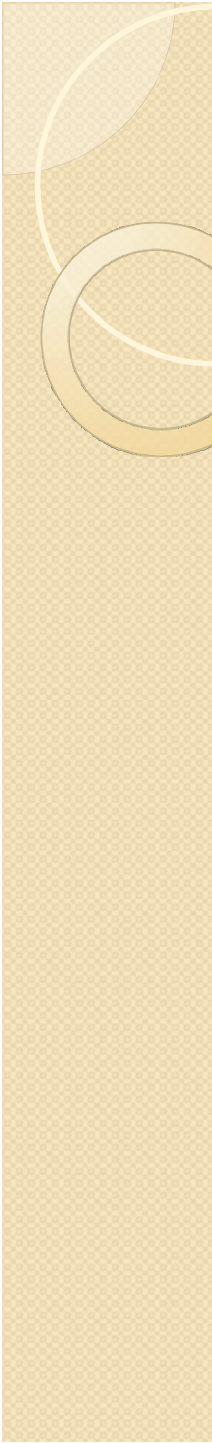
Not Important

It gives different levels of importance.

Example:

Question:

Who built the model?

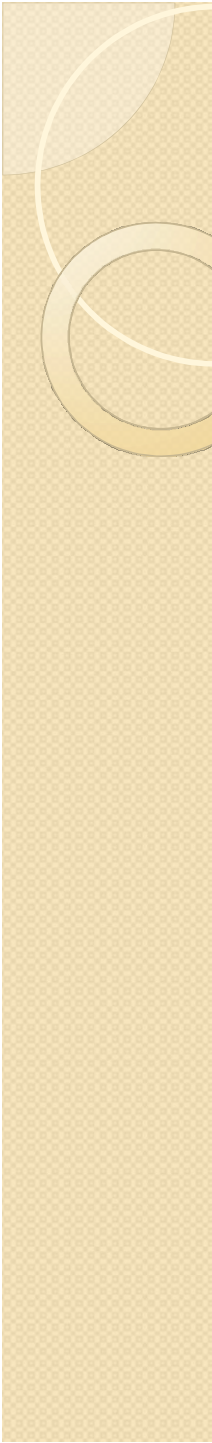


Word	Attention
The	0.02
AI	0.30
Engineer	0.65
built	0.10
model	0.05
dataset	0.03

Important

These are simplified example values.

Real models learn attention weights mathematically.



SLIDE 11 — From Attention to Self-Attention

Attention

Focus on relevant information.

Self-Attention

Every word can look at other words in the **same input**.

Example:

"The bank approved the loan."

The word:

bank

looks at:

- approved
- loan

This helps understand:

Bank = Financial institution

SLIDE 12 — Why Self-Attention Is Powerful

Consider:

"The animal didn't cross the road because it was too tired."

What does **it** refer to?

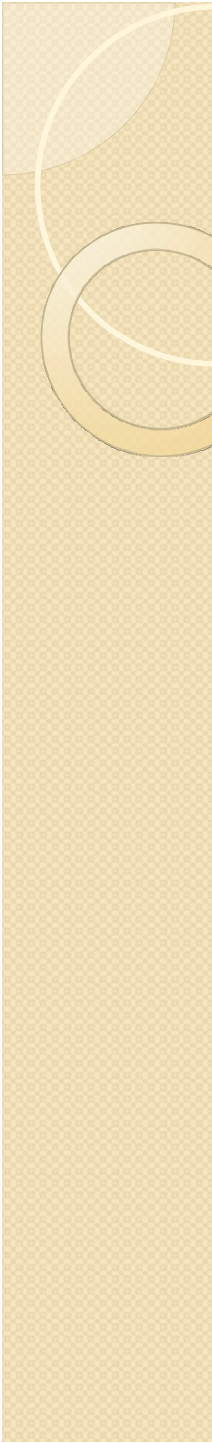
Answer:

The animal

The model needs to connect:

it —————→ animal

Self-Attention helps a word identify which other words are most relevant.



ACTIVITY 5 — What Does "It" Mean?

Display sentences.

Sentence 1

The robot stopped because **it** had no battery.

Q- What is "it"?

Sentence 2

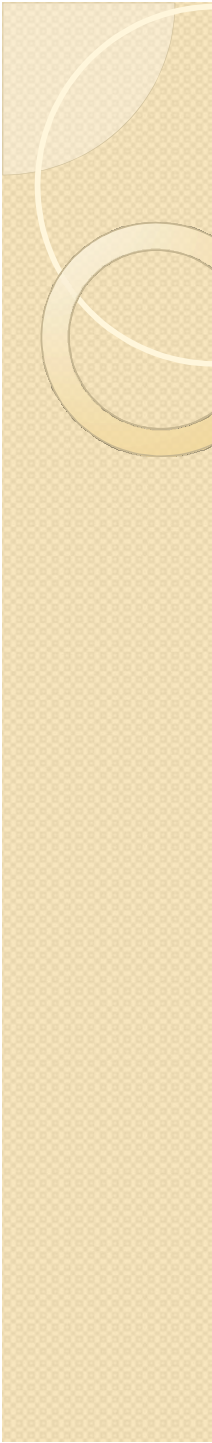
The computer processed the image because **it** was high quality.

Q- What is "it"?

Sentence 3

The students submitted the project after **they** completed testing.

Q- Who are "they"?



it $\rightarrow$ Robot

it $\rightarrow$ Image

they $\rightarrow$ Students

**Understanding these relationships is a language problem.
Self-Attention helps models learn them.**

SLIDE 13 — Self-Attention Visualization

Sentence:

The cat sat on the mat

Focus on:

cat

It may pay attention to:

The	cat	sat	on	the	mat
0.1	0.4	0.3	0.05	0.05	0.1

Focus on:

mat

Attention can be different.

The	cat	sat	on	the	mat
0.05	0.2	0.2	0.15	0.1	0.3

Key Idea

Every word can have a different attention pattern.

ACTIVITY 6 — Build an Attention Map

Sentence:-

The student solved the difficult AI problem.

A

Focus on:

student

Which words are most related?

B

Focus on:

solved

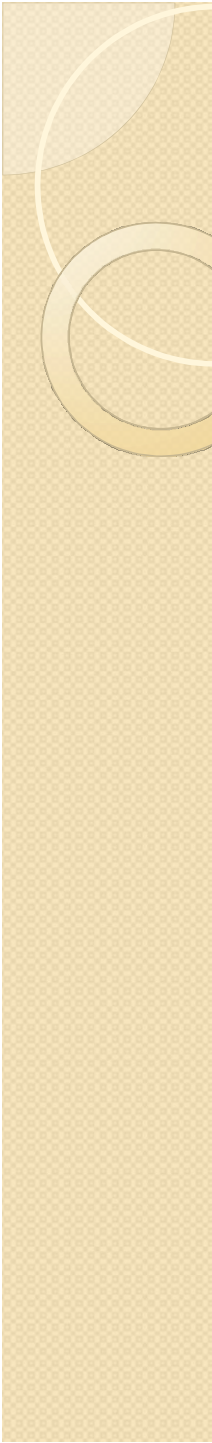
Which words are most related?

C

Focus on:

problem

Which words are most related?



Example:

student $\leftarrow$ solved

student $\leftarrow$ AI

problem $\leftarrow$ difficult

problem $\leftarrow$ AI

A real Transformer learns these relationships automatically.

SLIDE 14 — Attention vs Self-Attention

Attention	Self-Attention
Focuses on relevant information	Each token focuses on other tokens in the same input
Often relates one sequence/context to another	Relationships are learned within one sequence
Selects important information	Creates context-aware token representations

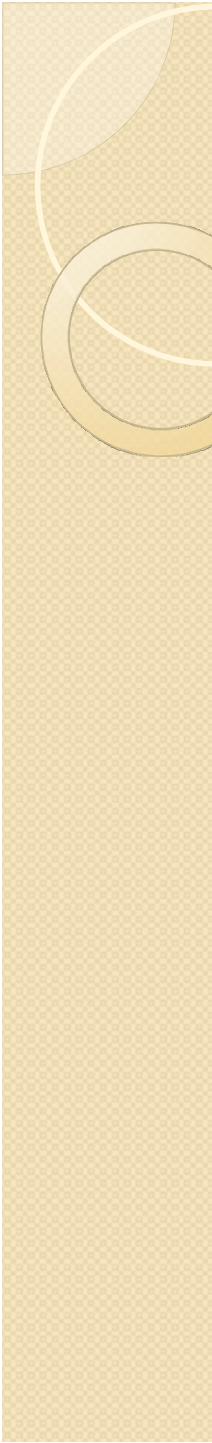
Simple Analogy

Attention:

"Look at the important page."

Self-Attention:

"Every word looks at the other words and decides which ones matter."



SLIDE 15 — The Transformer Idea

Stop Processing Everything Strictly One-by-One

Instead:

Input Tokens



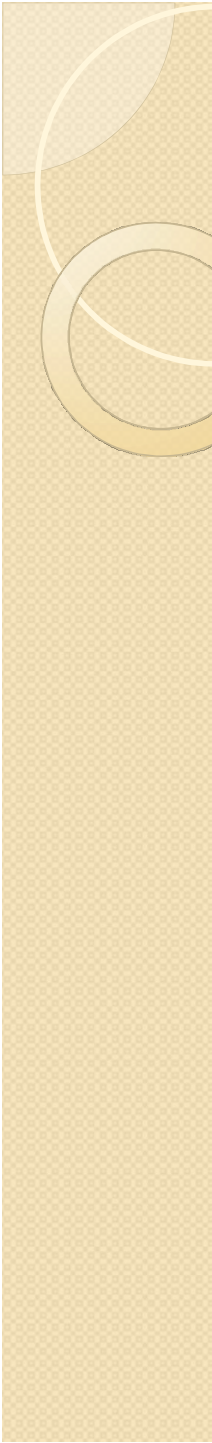
All Tokens Processed Together



Self-Attention Finds Relationships



Context-Aware Representations



Example:

I

love

learning

AI

A Transformer can compare relationships between tokens through attention mechanisms rather than passing information only step-by-step.

ACTIVITY 7 — Sequential vs Parallel Race

Students into two groups.

Group A — RNN

Give one student:

Artificial Intelligence is changing the world

They pass each word one by one.

1 → 2 → 3 → 4 → 5 → 6

Group B — Transformer

Give each student one word at the same time.

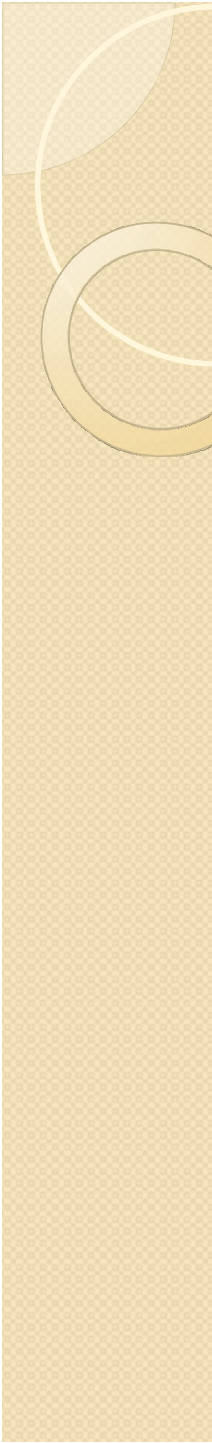
They can see the complete sentence and identify relationships.

Ask:

Which approach has more direct access to the full sentence?

Answer

Transformer-style self-attention



SLIDE 16 — Why Transformers Changed Everything

1. Better Long-Range Relationships

A word can directly relate to distant words.

2. More Parallelizable Computation

Tokens can be processed together during training.

3. Scales Better

Transformers can be trained with:

More data

More parameters

More computing power

4. Works Across Many Modalities

Transformers are used for:

Text

Images

Audio

Video

Multimodal AI

SLIDE 17 — RNN vs Transformer

Feature	RNN	Transformer
Processing	Sequential	Parallelizable
Long-distance context	More difficult	Direct attention connections
Training speed	Less parallelizable	Highly parallelizable on modern hardware
Main mechanism	Recurrence / hidden state	Self-Attention
Scalability	More limited for large language modeling	Foundation of modern LLMs

RNN

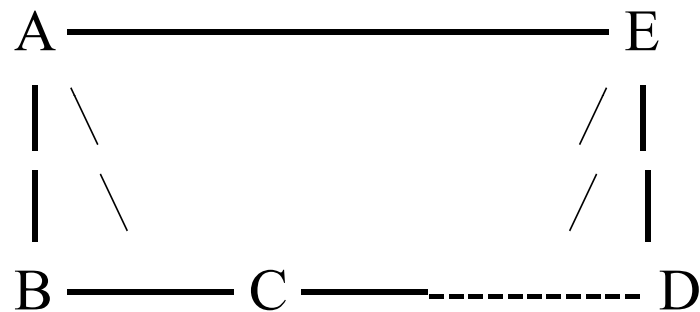
RNN

$$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E$$

To connect A and E:

$$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E$$

Transformer



Tokens can build relationships through attention.

Important

This is a simplified visualization, not the complete mathematical architecture.

SLIDE 19 — From Transformers to LLMs

Transformer Architecture



Train on Huge Amounts of Data



Billions of Parameters



Large Language Model



ChatGPT / Gemini / Claude / etc.

Examples of Transformer-Based Models

- GPT
- BERT
- T5
- Gemini
- Claude

SLIDE 20 — Real-World Connection

Example: ChatGPT

You type:

Explain Machine Learning with a student example.

Simplified process:

Prompt



Tokenization



Transformer Layers



Self-Attention



Context Understanding



Next Token Prediction



Response

ACTIVITY 8 — Become a Transformer

Sentence:

"The BCA student studied Transformers because the technology powers modern AI models."

Assign one student to each important word:

- BCA
- student
- studied
- Transformers
- technology
- powers
- AI
- models

If your word is "Transformers", which other words matter most?

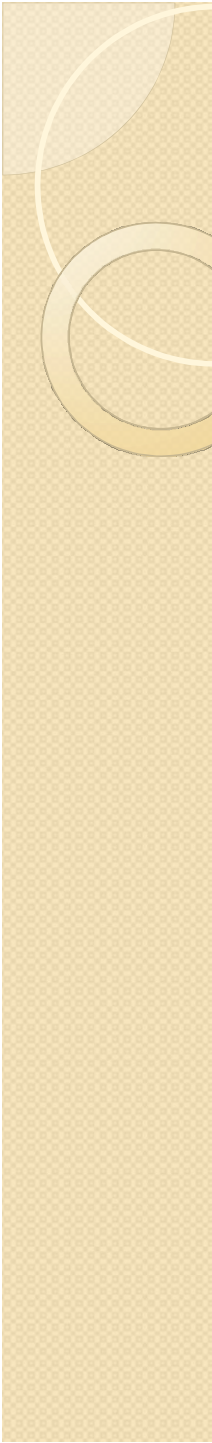
Likely:

- studied
- technology
- powers
- AI
- models

Students draw arrows between themselves.

Explain

This classroom network is a simplified representation of how self-attention connects information.



SLIDE 21 — Knowledge Check

1. Why is RNN sequential?

It processes information step-by-step and uses previous hidden information.

2. What is a long-term dependency?

When information far apart in a sequence needs to be connected.

3. What is Attention?

A mechanism that helps the model focus more on relevant information.

4. What is Self-Attention?

Tokens in the same sequence can determine which other tokens are important.

5. Why did Transformers become important?

They model relationships effectively and are highly parallelizable, making large-scale training practical.

FINAL SLIDE — Key Takeaways

RNN



Sequential Processing Problem



Long-Term Dependency Problem



Attention



Self-Attention



Transformer



Large-Scale Training



LLMs



ChatGPT / Modern Generative AI

- 
- <https://poloclub.github.io/transformer-explainer/>
 - <https://bbycroft.net/llm>